

Landmarks: Do They Help?

Benchmarking Facial Landmark Streams in Visual Speech Recognition

[Author Names Redacted for Blind Review]

Abstract

Visual Speech Recognition (VSR) has been dominated by appearance-only pipelines that feed raw pixel frames into convolutional architectures. In this work we investigate whether facial landmarks — a compact, structured, lower-dimensional representation of lip geometry — carry independent phonetic signal and whether they can complement appearance features in a dual-stream fusion framework. We introduce a dual-stream architecture combining (i) an appearance stream of CNN-extracted frame features and (ii) a landmark stream of 50-point lip and chin coordinates processed through a Transformer-BiLSTM pipeline. Using the LRW word-level benchmark and a curated 100-word hard subset derived via CMUDICT scoring and K-Means clustering, we find that landmarks alone reach 62% validation accuracy on the 100-word subset and 42% on the full 500-word LRW vocabulary — a research gap not previously reported. Our appearance-only baseline achieves 75% on the same 100-word subset, and early fusion experiments (concatenation, gated fusion, cross-attention) yield up to 95% training accuracy, though overfitting remains an open challenge. These results confirm that landmarks carry non-trivial phonetic signal and motivate their use as a lightweight complementary modality in VSR systems.

1 Introduction

Visual Speech Recognition (VSR) — the task of recognising spoken words or sentences from lip motion alone — has applications in accessibility aids for deaf and hard-of-hearing individuals, audio-visual restoration, surveillance in acoustically compromised environments, and as an input-processing front-end for speech-understanding modules in robotics. The field has advanced substantially through the adoption of deep learning, with convolutional and recurrent architectures achieving strong results on established benchmarks such as LRW [1] and LRS2 [2].

Despite this progress, virtually all competitive VSR pipelines operate on a single modality: the raw pixel appearance of the lip region. An alternative, lower-dimensional representation — facial landmarks — has received comparatively little systematic study as a *standalone or complementary* modality. Landmarks are lightweight, geometry-focused, and capture the structural deformation of the lip across time. Whether this representation carries discriminative phonetic signal, and whether it can augment appearance-based models, are empirical questions that remain largely unanswered.

This paper addresses three specific research questions:

- (1) Do landmarks carry useful phonetic signal as a standalone modality?
- (2) What is the landmarks-only benchmark performance on LRW, an evaluation gap in prior work?
- (3) Can a lightweight, resource-efficient model be built by fusing landmark and appearance streams?

Our contributions are as follows. We present a systematic ablation of landmark-only pipeline components (MLP, BiLSTM, Transformer) and report the first benchmarked landmark-only accuracy on LRW at both the 100-word and 500-word scales. We introduce a dual-stream fusion framework and evaluate three fusion strategies — concatenation, gated fusion, and cross-attention — on the combined modalities.

2 Related Work

2.1 Appearance-Based VSR

Early VSR work used hand-crafted features such as DCT coefficients or AAM parameters [8]. Deep learning pipelines supplanted these, with 3D convolutions [3] and hybrid CNN-RNN architectures [4] becoming the de facto standard. The MC3D + ResNet + BiLSTM architecture introduced by Stafylakis and

Tzimiropoulos [3] remains a competitive backbone, as does the use of pretrained ResNet18 frontends applied per-frame with a temporal recurrent module.

2.2 Landmark-Based Approaches

Facial landmarks have been used as auxiliary alignment signals [10] and in gesture recognition [9], but their standalone phonetic discriminability in a deep learning VSR pipeline has not been benchmarked. MediaPipe [5] provides real-time, accurate landmark detection, making it practical for large-scale feature extraction. Our work fills this gap by treating landmarks as a first-class input stream.

2.3 Multi-Modal and Multi-Stream Fusion

Audio-visual speech recognition [7] uses complementary modalities to improve robustness. Within the visual domain, multi-stream fusion has been less explored. Cross-attention mechanisms for modality fusion [6] offer a principled way to allow each stream to attend to the other’s features. We apply these ideas to landmark–appearance fusion.

3 Datasets

3.1 LRW (Lip Reading in the Wild)

LRW [1] is a word-level benchmark containing over 500 word classes extracted from BBC broadcasts, with 1000 utterances per word. Each clip is 29 frames at 25 fps. The dataset is 70 GB in raw .mp4 format.

3.2 LRS2

LRS2 [2] is a sentence-level benchmark drawn from BBC programmes, containing over 140,000 variable-length clips with character-level transcriptions. It includes a Pretrain partition and a smaller Main partition used for fine-tuning. The dataset is 50 GB in raw format.

3.3 Hard-100 Subset Construction

To validate our landmark pipeline efficiently, we constructed a 100-word hard subset from LRW. Each word was scored using its CMU Pronouncing Dictionary [11] representation: words with fewer, more visually ambiguous phonemes received higher difficulty scores. K-Means clustering ($k = 2$) was applied to the resulting score distribution, which revealed two naturally separable clusters. The top 100 words of the harder cluster were selected as the Hard-100

evaluation set. This subset was used for all landmark ablation experiments before scaling to the full 500-word vocabulary.

4 Architecture

Our dual-stream system consists of an appearance stream, a landmark stream, and a fusion module. Figure ?? provides an overview.

4.1 Appearance Stream

The appearance stream follows the MC3D + ResNet18 + BiLSTM paradigm. Input is a sequence of grayscale frames cropped to the lip region (96×96 pixels). A single heavy 3D convolutional frontend (Conv3D, kernel $5 \times 7 \times 7$, 64 channels, GroupNorm, ReLU, MaxPool3D) extracts spatiotemporal features. The time dimension is then collapsed into the batch axis so that a pretrained ResNet18 — with its first convolutional layer replaced to accept 64-channel input and with MaxPool removed — processes each frame independently. An AdaptiveAvgPool layer aggregates spatial features, and a 2-layer BiLSTM (256 hidden units per direction, dropout 0.3) models temporal dynamics. For the appearance-only baseline, a linear classifier maps BiLSTM hidden states to class logits.

4.2 Landmark Stream

The landmark stream takes as input a sequence of 50 2D keypoints localised to the lip and chin region by MediaPipe Face Mesh [5]. Each frame is represented as a 100-dimensional vector (50 (x, y) pairs), normalised to the bounding box of the detected face.

The landmark encoder is a Transformer–BiLSTM pipeline. A positional encoding is added to the input sequence, followed by a multi-head Transformer encoder (depth, attention heads, and feedforward dimensions determined by ablation; see Section 6). The Transformer output is passed to a 2-layer BiLSTM, and the resulting hidden states are used as the landmark representation. For standalone evaluation, a linear classifier is applied.

4.3 Fusion Module

Three fusion strategies are evaluated:

Concatenation. BiLSTM hidden states from both streams are concatenated and passed through a linear projection to the classifier.

Gated Fusion. A learnable gating vector $g = \sigma(W_g[h_a; h_l] + b_g)$ modulates the contribution of each stream, where h_a and h_l are appearance and landmark representations respectively.

Cross-Attention. The appearance representation attends to the landmark representation (and vice versa) using scaled dot-product attention [6], allowing the model to selectively weight landmark features conditioned on visual context.

5 Experimental Setup

5.1 Training Details

All experiments were conducted on Google Colab using an NVIDIA T4 GPU (16 GB VRAM). The physical batch size was constrained to 2 due to VRAM limits; 8-step gradient accumulation brought the effective batch size to 16. PyTorch Automatic Mixed Precision (AMP) with a GradScaler was used throughout. The Adam optimiser was used with a learning rate scheduler; specific LR values and patience settings varied per experiment and are detailed in Section 6.

5.2 CTC Training (LRS2)

For the sentence-level LRS2 stream, Connectionist Temporal Classification (CTC) loss was used with a 39-class character alphabet. Curriculum learning was applied: training began on clips with transcripts of ≤ 50 characters, expanded to ≤ 100 characters, then to the full dataset. Blank-collapse prevention heuristics included a loss ceiling (batches with loss > 1000 were skipped) and a CTC alignment guard (batches where $2L_{\text{target}} + 10 \geq L_{\text{video}}$ were skipped). A three-mode smart resume system (`latest`, `best`, `none`) was implemented to handle Colab session disconnects.

5.3 Data Engineering

The raw LRW dataset (70 GB, .mp4) was converted to NumPy arrays to eliminate per-batch decompression overhead, resulting in approximately 140 GB on-disk but substantially faster data loading. The LRS2 dataset was processed on Google Cloud Platform (GCP) using a VM with an L4 GPU (8 vCPUs, 32 GB RAM) and Blob Storage, motivated by similar CPU-bound decompression bottlenecks.

6 Ablation Study: Landmark Stream

6.1 Validation of Landmark Signal (Experiment 001)

Before building a full pipeline, we asked whether landmarks carry any phonetic signal at all. A simple 4-layer Transformer encoder was trained directly on the raw 50-point sequences on the Hard-100 subset. This architecture contains no convolutional component and performs no feature extraction beyond positional encoding. It achieved 61.64% validation accuracy, confirming that landmarks carry sufficient signal to discriminate 100 word classes above chance (random baseline: 1%). This result justified continued investment in the landmark stream.

6.2 Classifier Depth: MLP Baselines (Experiments 002–003)

MLPs of varied depths (2, 3, and 4 hidden layers) were evaluated as classifiers on raw landmark features. Deeper MLPs improved training accuracy but showed limited generalisation, motivating the addition of temporal modelling.

6.3 Adding Temporal Context: BiLSTM (Experiment 004)

Appending a 2-layer BiLSTM between the feature encoder and the MLP classifier produced a measurable improvement in validation accuracy over MLP-only baselines, confirming the importance of modelling temporal dynamics in the landmark sequence.

6.4 Transformer Integration (Experiments 005–006)

Replacing or prepending a Transformer encoder (4 layers, 4 attention heads) before the BiLSTM produced the best results in this group. Experiment 006 established the first complete Transformer–BiLSTM landmark pipeline.

6.5 Augmentation (Experiments 007–008)

Coordinate jitter and random temporal subsampling were evaluated as augmentation strategies. Moderate jitter improved generalisation slightly; aggressive subsampling degraded performance, suggesting the model relies on dense temporal coverage of lip motion.

6.6 Architecture Granularity (Experiments 009–014)

A grid search over Transformer depth (2, 4, 6 layers), positional encoding variants (sinusoidal vs. learned), and output projection dimension (128, 256, 512) was conducted. Key findings:

- 4-layer Transformer depth was optimal; deeper models overfit on the Hard-100 subset.
- Sinusoidal positional encoding marginally outperformed learned encoding.
- 256-dimensional projections balanced capacity and generalisation.

Finalised Baseline (Experiment 010). The finalised landmark-only pipeline — 4-layer Transformer + 2-layer BiLSTM + linear classifier — achieved **62% validation accuracy** on the Hard-100 subset. This matches the initial 4-layer Transformer-only result from Experiment 001, suggesting the Hard-100 task has an architecture-agnostic ceiling near 62%, likely imposed by visual ambiguity in the landmark representation itself rather than model capacity.

6.7 Scaling to Full LRW Vocabulary (Experiments 015–018)

The finalised pipeline was extended to the full 500-word LRW vocabulary, with and without warm-starting from Hard-100 checkpoint weights.

Table 1: Landmark-only accuracy on LRW.

Vocabulary	Warm-start	Val Acc (%)
Hard-100	–	62.0
LRW-500	No	39.8
LRW-500	Yes	42.0

Warm-starting from the Hard-100 checkpoint provided a consistent improvement of approximately 2 percentage points over training from scratch on LRW-500, confirming positive transfer from the curated subset. The final **42% landmark-only accuracy on the full LRW-500 benchmark** constitutes, to our knowledge, the first reported result of this kind in the literature.

7 Appearance Stream Results

The appearance-only baseline (MC3D + ResNet18 + BiLSTM, trained on LRW) achieved **75% validation accuracy** on the Hard-100 subset, establishing the

appearance stream as the dominant modality by a margin of 13 percentage points over landmarks alone.

For the sentence-level LRS2 stream, CTC-based training on the MC3D + ResNet18 + BiLSTM architecture reached a best validation loss of **2.608** after curriculum pretraining and **2.674** after partial fine-tuning on LRS2 Main. Decoder output at this stage was limited to the three most visually dominant viseme classes (I/E, U/O, and the high-frequency token I), consistent with partial viseme collapse — a known threshold phenomenon that requires val loss < 2.5 and substantially more compute to overcome.

8 Fusion Results

Table 2: Fusion strategy results on Hard-100 (features post-BiLSTM).

Fusion Strategy	Train Acc (%)	Val Acc (%)
Concatenation	93.1	–
Gated Fusion	94.3	–
Cross-Attention	95.1	–

All three fusion strategies achieved high training accuracy (up to 95.1%). However, all models exhibited significant overfitting, with validation accuracy failing to match training accuracy. We attribute this to extracting fusion features *after* the BiLSTM: at this stage the temporal representations from each stream are highly compressed and may share redundant structure, causing the fusion module to memorise training patterns rather than learn generalisable cross-modal dependencies.

A natural next step — currently in progress — is to perform fusion on features extracted *before* the BiLSTM, where richer temporal structure remains present in both streams and the fusion module has more varied signal to work with.

9 Discussion

9.1 Do Landmarks Help?

The evidence from our experiments points to a qualified *yes*. Landmarks carry independent phonetic signal (62% on Hard-100, 42% on LRW-500 standalone), and early fusion experiments show high training-time benefit. The unresolved question is whether this signal generalises after fusion, which the overfitting of post-BiLSTM fusion features currently prevents us from confirming. Pre-BiLSTM fusion remains the most promising path forward.

9.2 Architecture-Agnostic Ceiling on Landmarks

The near-identical validation accuracy of the 4-layer Transformer-only model (61.64%, Experiment 001) and the full Transformer-BiLSTM pipeline (62.0%, Experiment 010) on Hard-100 suggests that the ceiling is not architectural but representational. The 50-point landmark set, despite covering the lip and chin region, may lack the expressiveness to distinguish consonant classes that differ primarily in fine-grained spatiotemporal patterns below the resolution of sparse keypoints. Denser landmark sets or auxiliary geometric features (e.g., lip curvature, inter-landmark velocity) may push this ceiling higher.

9.3 Lightweight Design

A secondary goal was to build a resource-efficient model. The landmark stream is substantially lighter than the appearance stream: it takes 100-dimensional inputs versus full-resolution image tensors, requires no convolutional frontend, and runs on CPU-extracted MediaPipe keypoints. Landmark-only inference therefore incurs a fraction of the compute cost of appearance-based models, making it attractive for edge deployment or as a low-power fallback modality.

9.4 Resource Constraints

All experiments were conducted under significant compute constraints. Free-tier Colab with a T4 GPU (16 GB VRAM) forced a physical batch size of 2 throughout, introducing BatchNorm instability in the appearance stream and limiting iteration speed on LRS2 (approximately 16 minutes per epoch). Published LRS2 results were achieved on multi-GPU A100/V100 clusters with batch sizes of 16–32. The architecture implemented here is equivalent to competitive published approaches; the gap in LRS2 final performance is attributable entirely to compute budget rather than architectural choice.

10 Conclusion

We presented a systematic study of facial landmarks as a modality for Visual Speech Recognition. Our findings demonstrate that:

- (1) A 4-layer Transformer on 50-point lip landmarks achieves 61.64% on a curated Hard-100 LRW subset, confirming non-trivial phonetic signal.
- (2) A finalised Transformer-BiLSTM landmark pipeline reaches 62% on Hard-100 and **42% on**

the full LRW-500 benchmark — the first reported landmark-only result on this dataset.

- (3) The appearance-only baseline achieves 75% on Hard-100, confirming appearance as the dominant modality.
- (4) Early fusion (concatenation, gated, cross-attention) reaches 95% training accuracy but overfits, motivating pre-BiLSTM fusion as future work.

The landmark stream is a lightweight, complementary modality that carries genuine phonetic signal. Whether that signal provides a consistent held-out accuracy benefit in fusion — rather than only a training-time benefit — remains the central open question of this line of work.

Acknowledgements

Experiments were conducted using Google Colab and Google Cloud Platform. Landmark extraction used MediaPipe Face Mesh. The LRW and LRS2 datasets are used under their respective academic licences.

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